

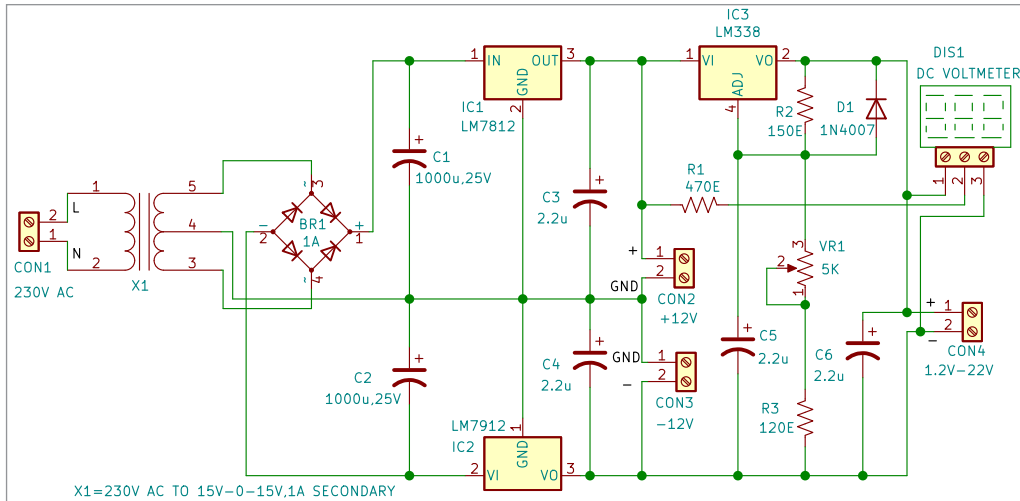


Fig. 3: Circuit diagram

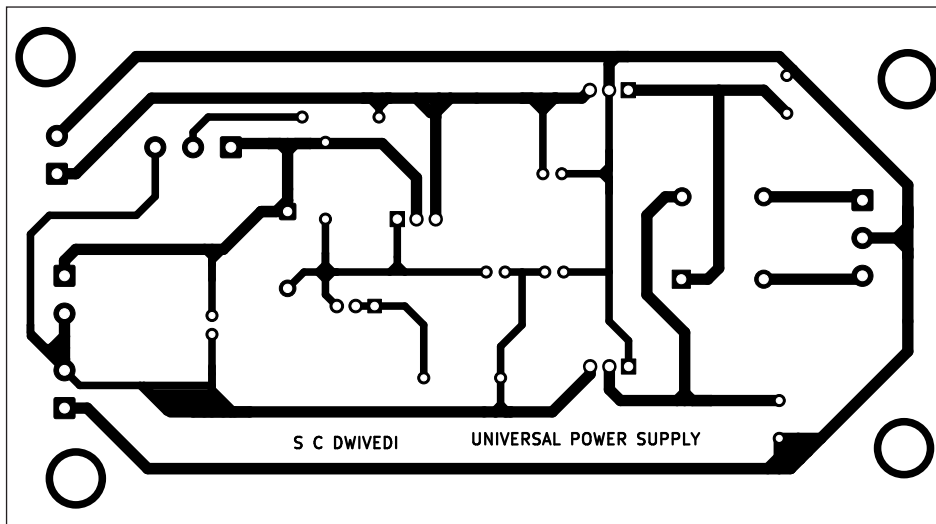


Fig. 4: Actual sized PCB

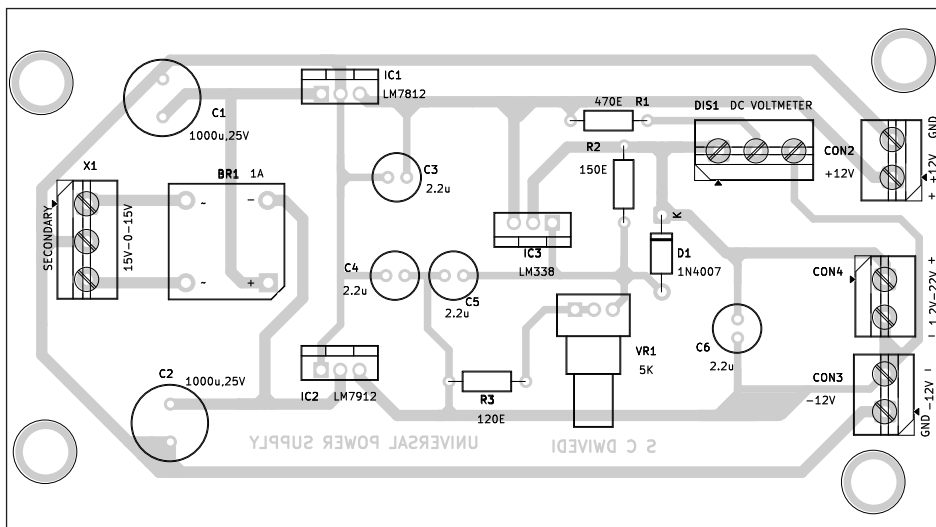


Fig. 5: Components layout

Capacitors C1 and C2 work as filters to smooth the rectified voltage and reduce ripple. After filtering, the circuit develops approximately +21V DC and -21V DC before regulation.

The positive regulated supply section is built around IC1 (LM7812), which provides a stable +12V regulated output. Capacitors C3 and C4 improve regulator stability and reduce output noise

and oscillations. The regulated +12V output is available at connector CON2 and can be used for powering microcontrollers, digital circuits, relay boards, and logic devices.

The negative regulated supply section uses IC2 (LM7912), which generates a regulated -12V output. This negative supply is useful in operational amplifier circuits, analogue signal-processing systems, instrumentation circuits, and audio applications requiring dual power rails. The regulated -12V output is available at connector CON3.

The adjustable output section is built around IC3 (LM338), an adjustable voltage regulator. The output voltage is controlled using resistors R2 and R3 and potentiometer VR1. Diode D1 protects the regulator from reverse discharge current generated by the output capacitor during sudden power-off conditions.

By adjusting preset VR1, the output voltage can be varied smoothly from nearly 1.2V up to about 22V DC, depending on the transformer rating and load conditions. Capacitors C5 and C6 improve output stability and transient response. A DC voltmeter module (DIS1) is